

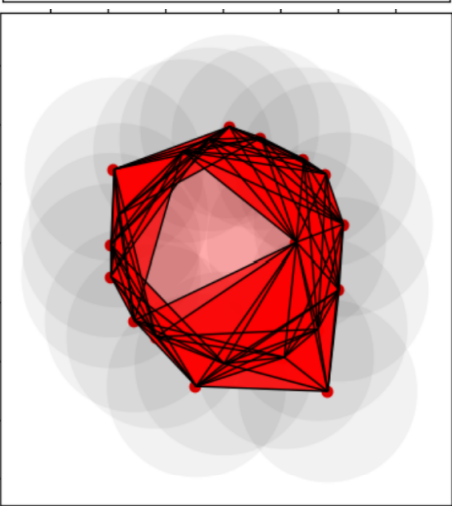
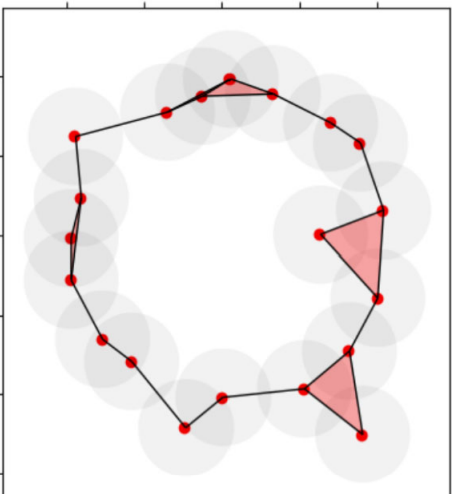
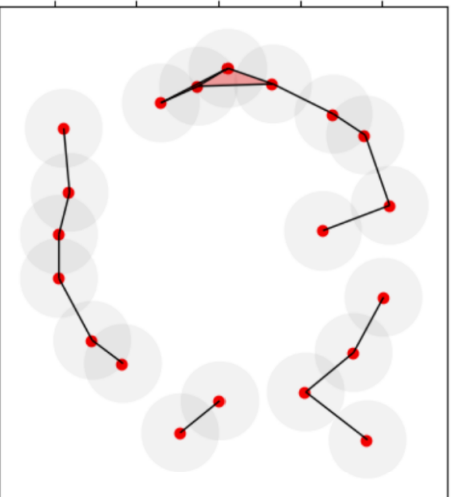
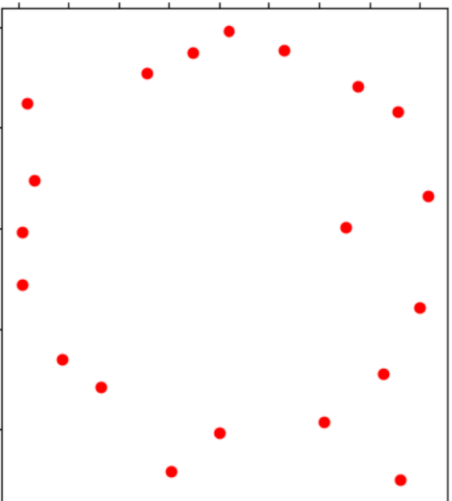
BOX FILTRATION AND BOX MAPPER

Bala Krishnamoorthy
Washington State University

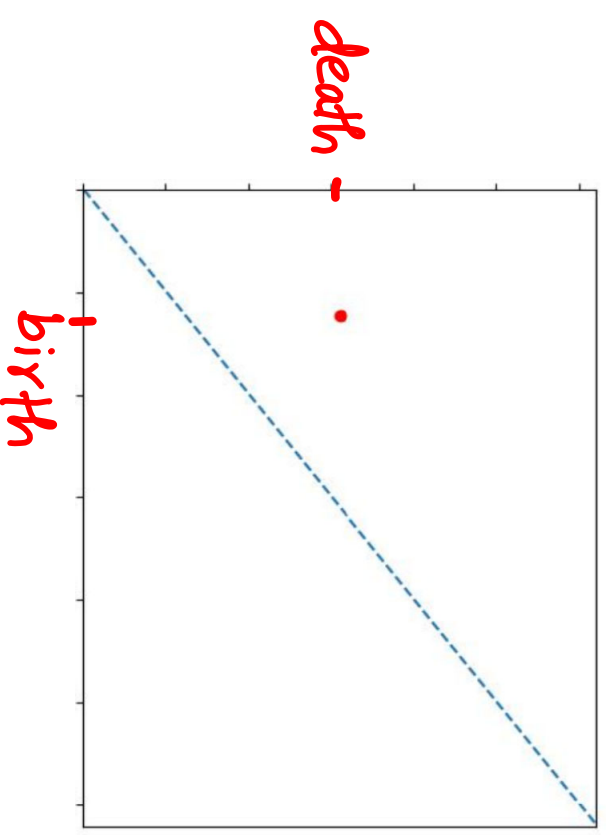
Rashant Gupta, Enrique Alvarado

JACT, 2025 (arXiv: 2404.05859)

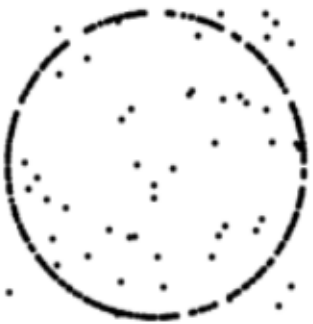
PERSISTENT HOMOLOGY



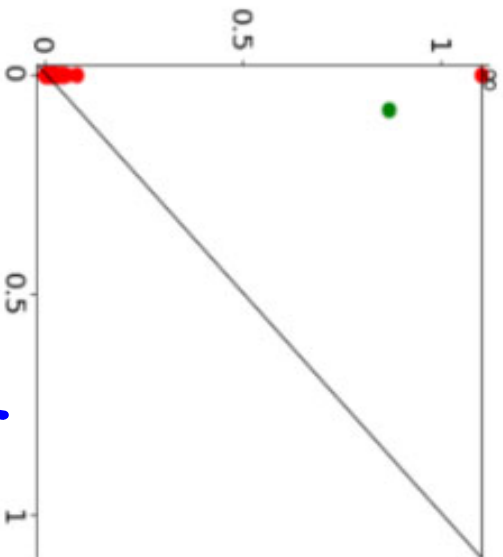
Vietoris-Rips (VR) filtration



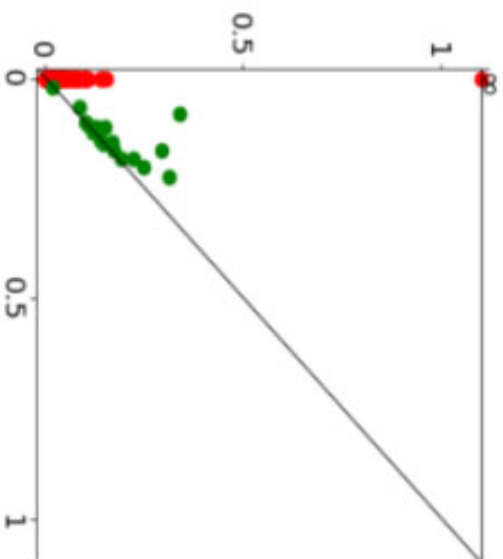
FILTRATION AND NOISE/OUTLIERS



points from circle +
some random points



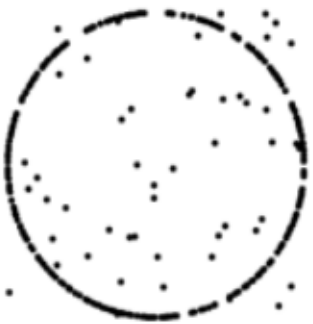
VR: circle



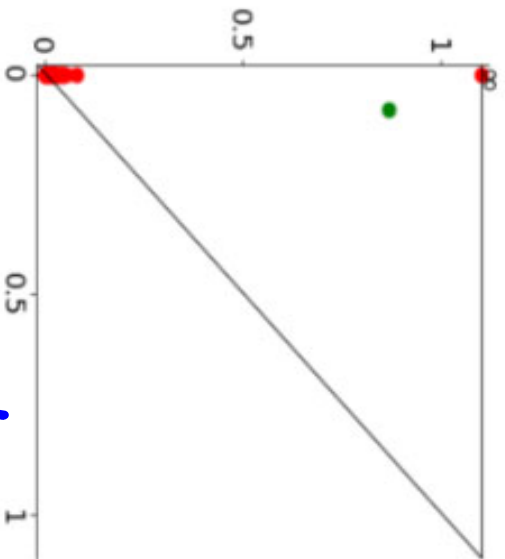
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FILTRATION AND NOISE/OUTLIERS

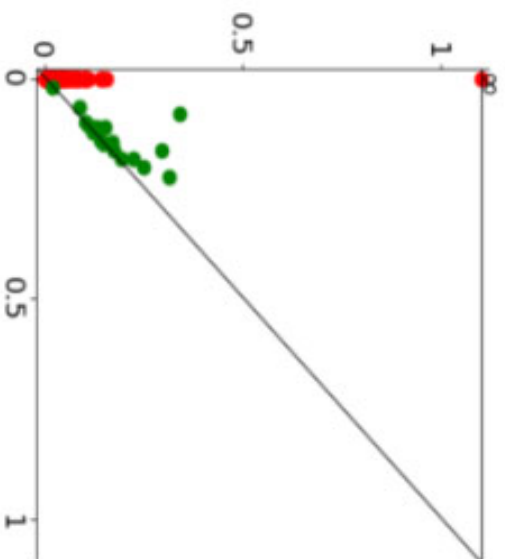
* Anai et al.(2019): DTM-based Filtrations



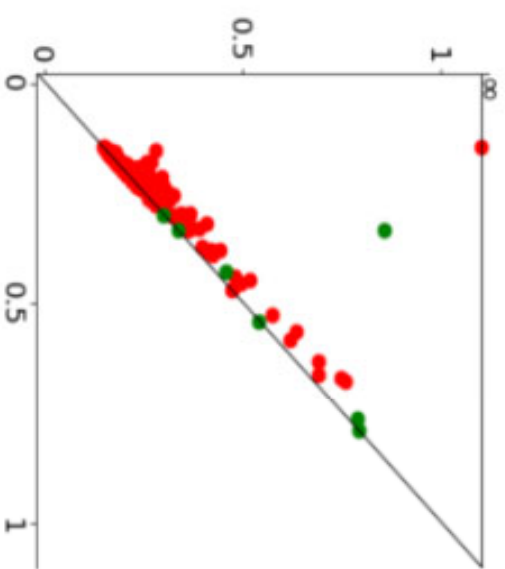
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VR: circle



VR: circle+noise



DTM: circle+noise

RELATED WORK

caution: not exhaustive!

* DTM-based Filtrations: Anai et al. (2019)

— Robust PH for measures: Buchet et al. (2016)

— C.I.s for PH on measure metric spaces:

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— Witnessed k -distances: Guibas et al. (2011)

RELATED WORK

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- * kernel density estimation
- topological consistency: Bobrowski et al. (2017)
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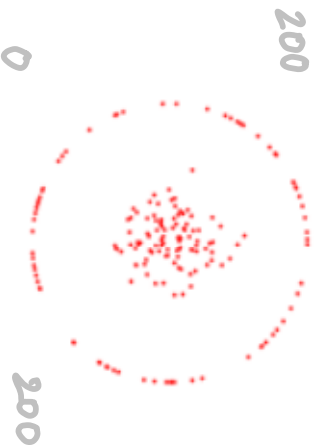
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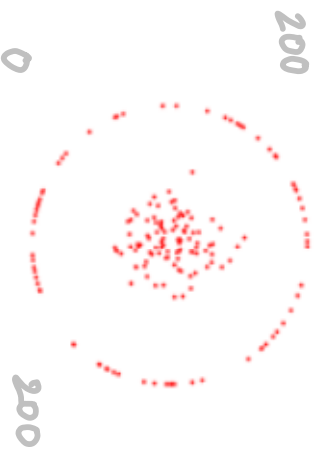
! All approaches grow balls centered @ each point uniformly in all directions

FILTRATION: HARDER INSTANCE



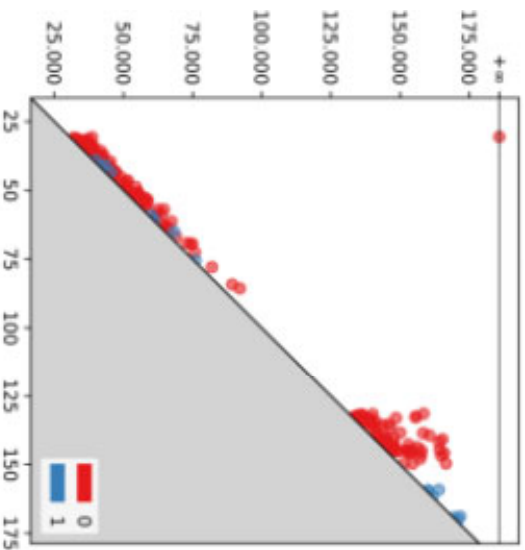
points from circle +
points in middle cluster

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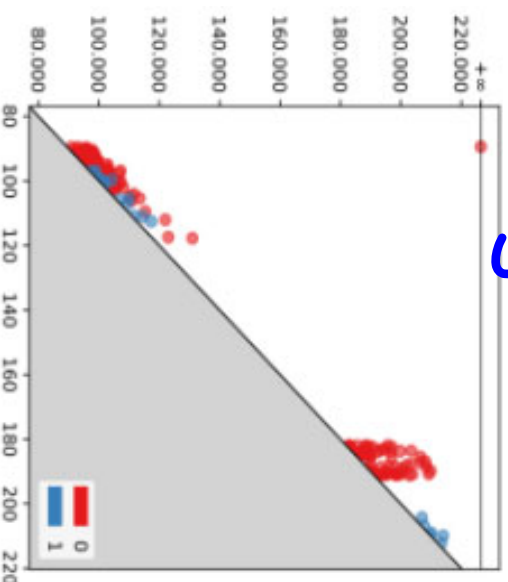


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DTM fails to identify circle

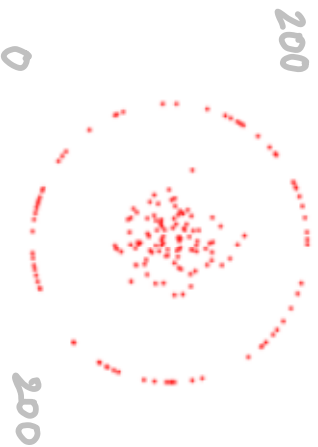


DTM, $m=0.3$



DTM, $m=0.7$

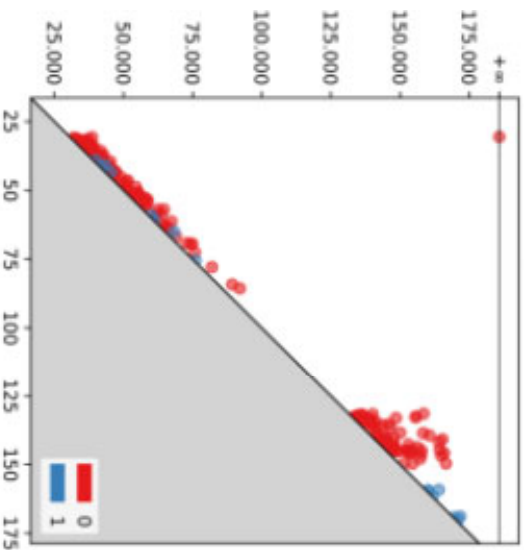
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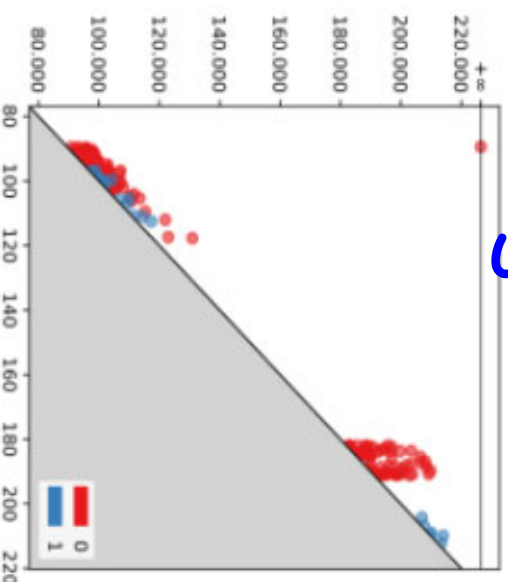
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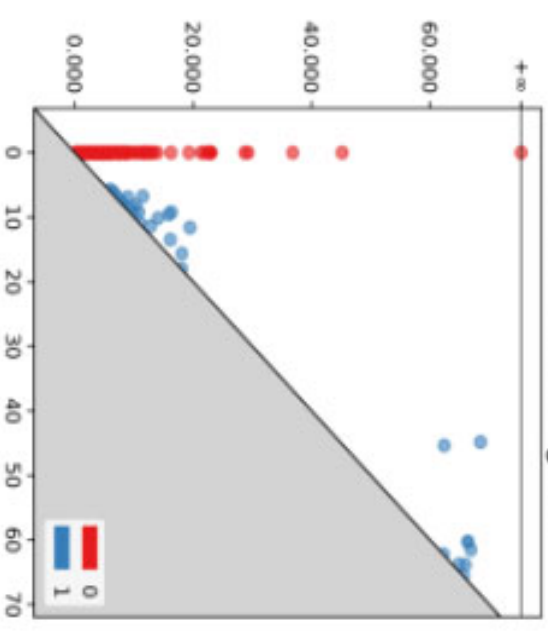
VR does better !



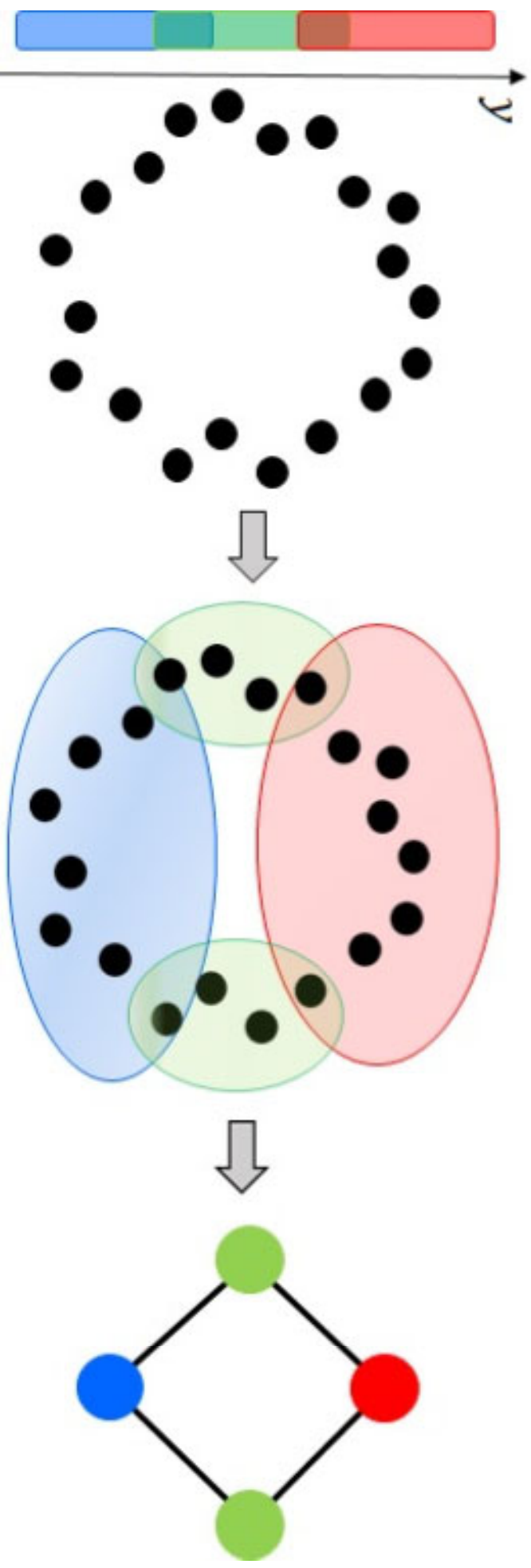
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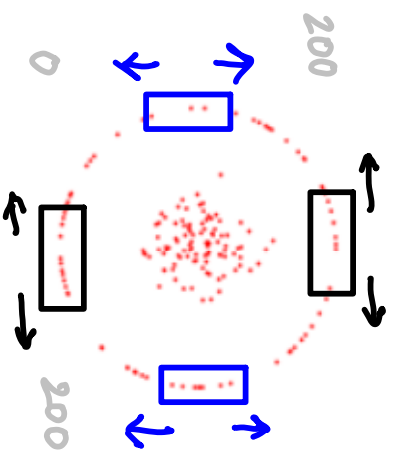
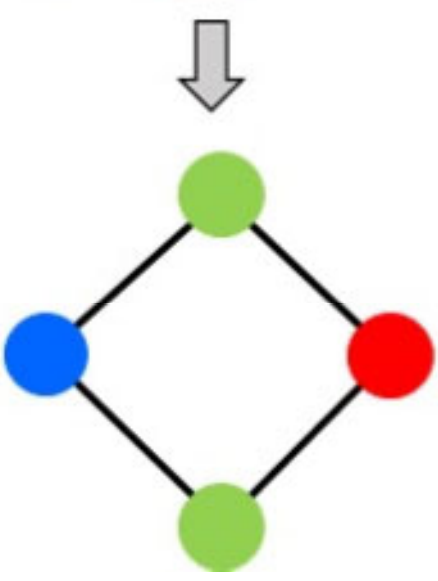
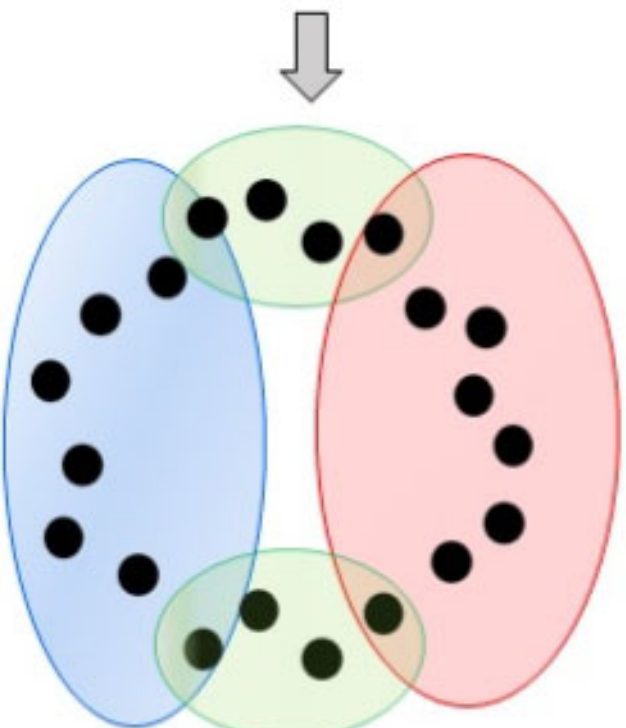
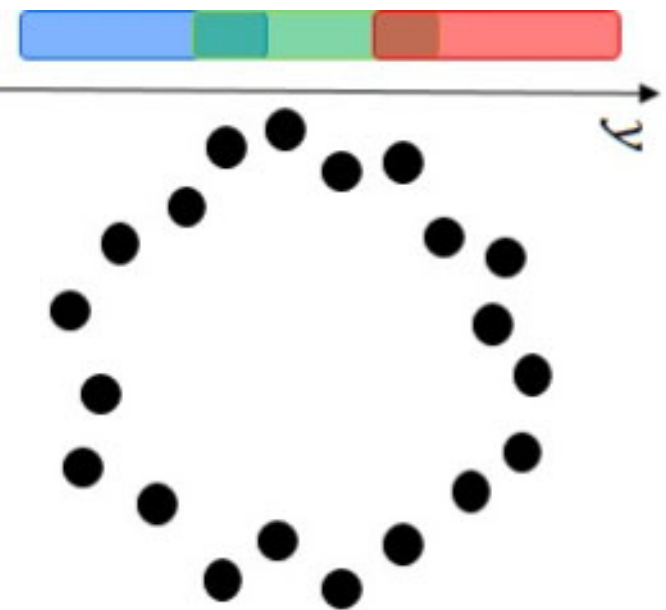
DTM, $m=0.7$



MAPPER



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QUESTIONS

? Build filtration by growing boxes non-symmetrically?

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- ? Stability?
- ? Unify filtration and mapper frameworks?

RESULTS

For finite point set X in $\mathbb{R}^d$

- ✓ Filtration using boxes (hyper-rectangles)

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 - compute filtration by solving sequence of LPs

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- ✓ Filtration using boxes (hyper-rectangles)
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- ✓ stability [$d_b \leq f(\text{Gromov-Hausdorff})$]

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For finite point set X in $\mathbb{R}^d$

- ✓ – point cover (start w/ one box for each $x \in X$)
- pixel cover (start w/ pixelization of space)

RESULTS

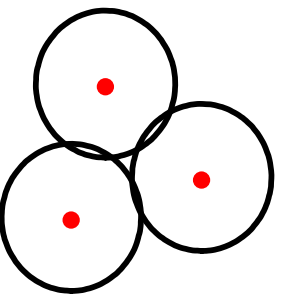
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- ✓ every box (point or pixel) cover gives a mapper
 - a mapper framework w/ stability guarantees

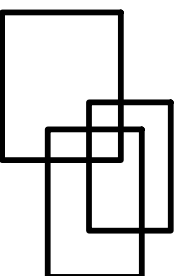
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- ✓ "box $\check{C}_{ech}$ " $\equiv$ "box V_R "



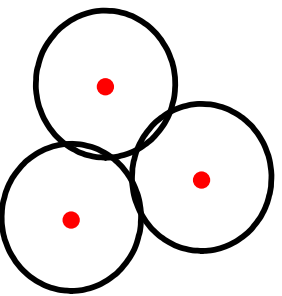
$\check{C}_{ech} \subset V_R$



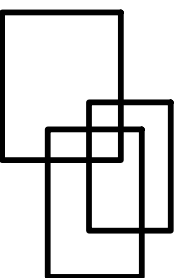
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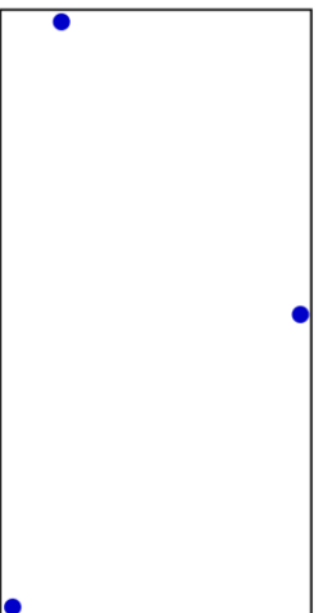
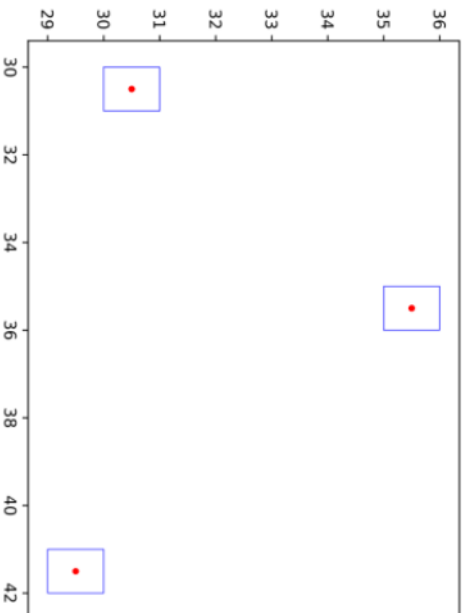


check only
pairwise intersections!

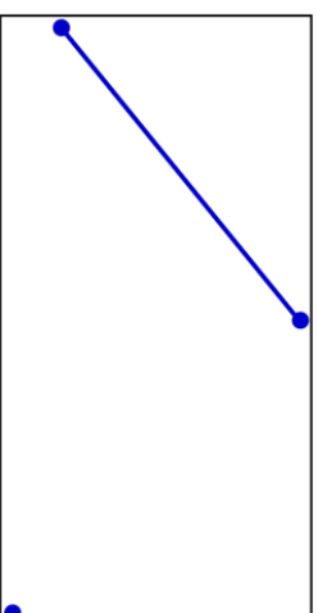
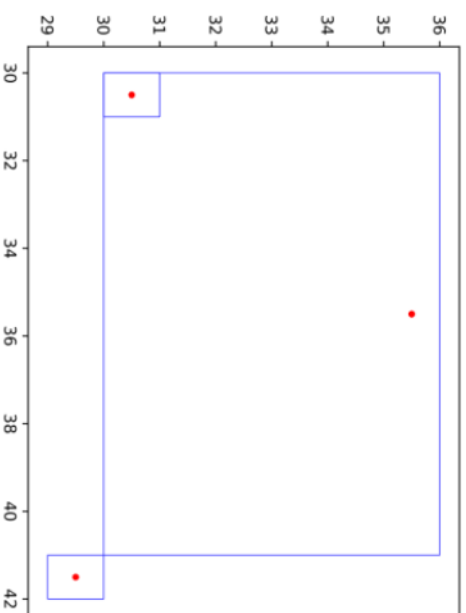
Box Filtration: Illustration

$$\alpha = 0.6$$

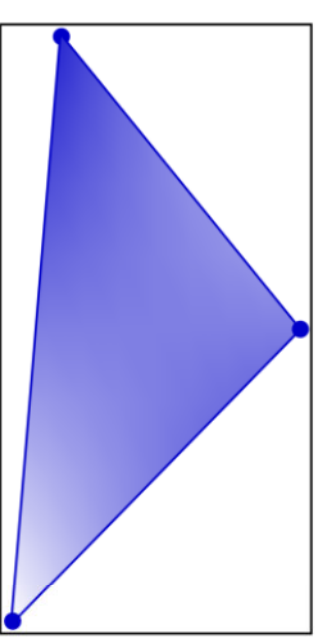
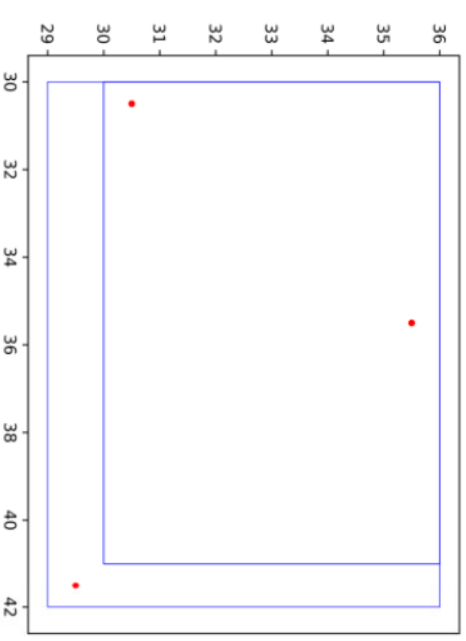
$$\pi = 4$$



$$\pi = 11$$



$$\pi = 12$$



Box Filtration: Cost Function

* Cost of box B over n 'th $N \supseteq B$

$$\mathcal{L}(B) = \alpha \sum_{x \in X \cap N} \text{dist}_\infty(x, B) + (1-\alpha) |B|_1, \quad \alpha \in [0, 1]$$

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L_1 diameter of box

— $\alpha = 1$: "Čech with boxes"

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* Expansion of B over $N := \pm \pi$ enlargements of B

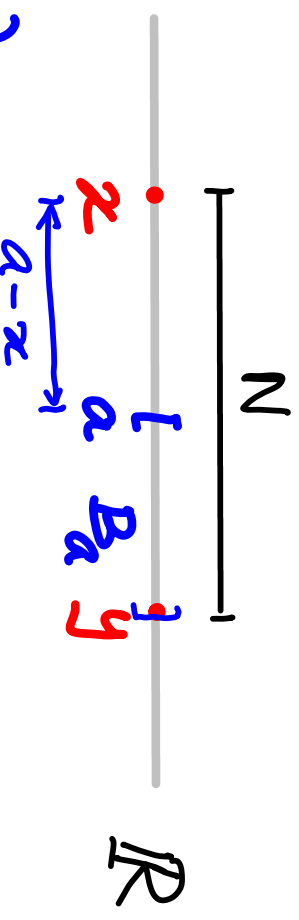
minimize $\mathcal{C}(\tilde{B})$

s.t. $B \subseteq \tilde{B} \subseteq N$

COST FUNCTION: PROPERTIES

* Proposition minimizer may not be unique

$$X = \{x, y\}, B = \{y\}$$

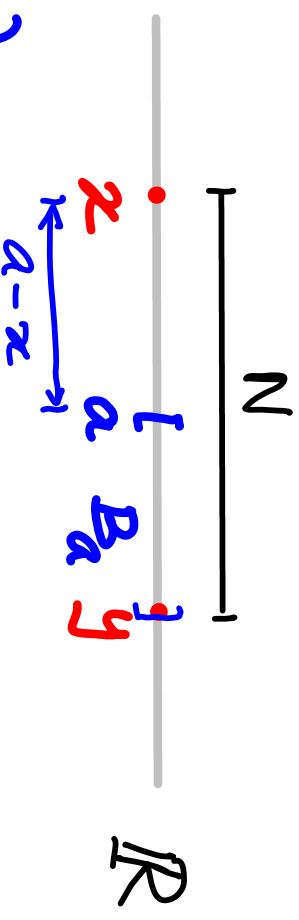


$$\mathcal{L}(B_a) = \alpha(a-x) + (1-\alpha)(y-a)$$

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$$X = \{x, y\}, B = \{y\}$$



$$\mathcal{L}(B_a) = \alpha(a-x) + (1-\alpha)(y-a)$$

$\Rightarrow$ for $\alpha = \frac{1}{2}$, B_a is minimizer for any $x \leq a \leq y$

COST FUNCTION: PROPERTIES

* Theorem If $B_1, B_2 \in \text{Sol}_\alpha(B, N)$ then
 $B_1 \cup B_2 \in \text{Sol}_\alpha(B, N)$.
↳ optimal solutions for
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- * find largest $B' \in \text{Sol}_\alpha(B, N)$
 - expand $N' := N \pm \pi$, search $\text{Sol}_\alpha(B', N')$
- * guarantees $B \subseteq B'$
⇒ filtration using boxes!

Box FILTRATION: ALGORITHM

* start with $\{B_{j_0}, \dots, B_{p,0}\}$ p pivot boxes

* iteration j :

for $k = 1 \dots p$

$B_{k,j} := \text{maximal box in } \text{Sol}_\alpha(B_{k,j-1}, B_{k,0} \pm j\pi)$

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* solve LP for each $B_{k,j}$

— always choose maximal box

STABILITY

* Assumption on $X \subseteq \mathbb{R}^d$:

- # points from X on any facet of box $B \leq c$
- small change in $|B|_1 \iff$ small change in $|\{x \in B\}|$ constant $\nearrow$

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- similar to general position (for balls)

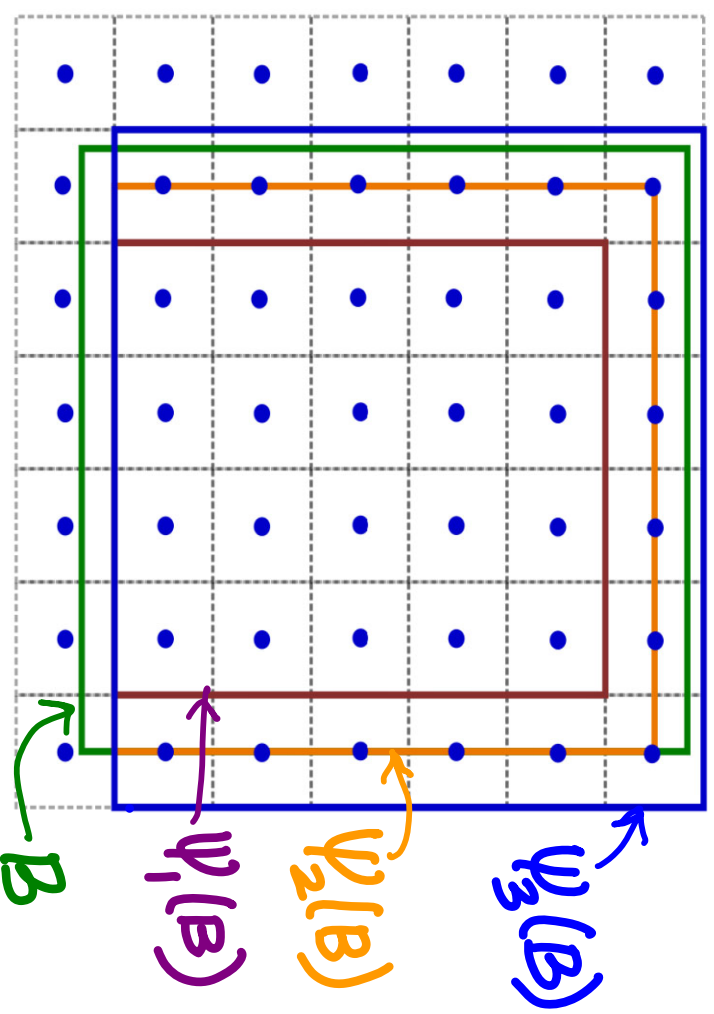
* Theorem If $\bar{U}, \bar{V}$ are persistence modules using box filtration for X, Y with $2d_{GH}(X, Y) < \epsilon$.
Then $d_b(d_{gm}(\bar{U}), d_{gm}(\bar{V})) \leq \gamma$ where
 $\gamma = \max \{f(\alpha, c, \epsilon), 2\epsilon\}$

PIXEL COVER

- * Use pixelization of space containing X
 - each pixel is unit cube w/ integer vertices
 - $\pi \in \mathbb{Z}$

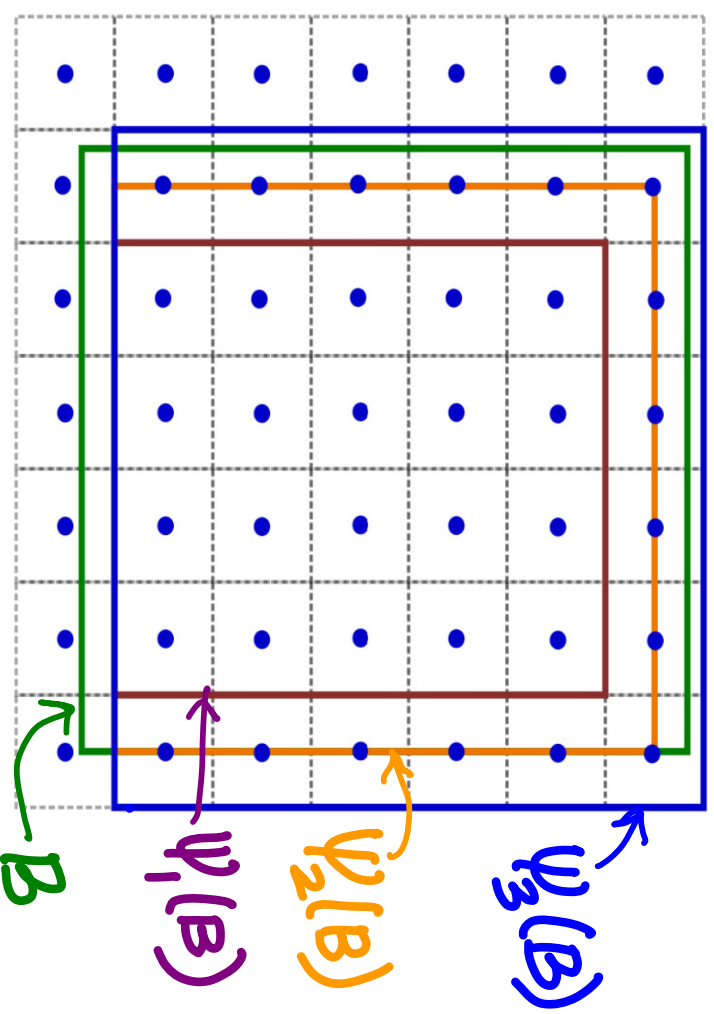
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- * round box B to fit the pixel space $\psi_i(B)$, $i=1,2,3$



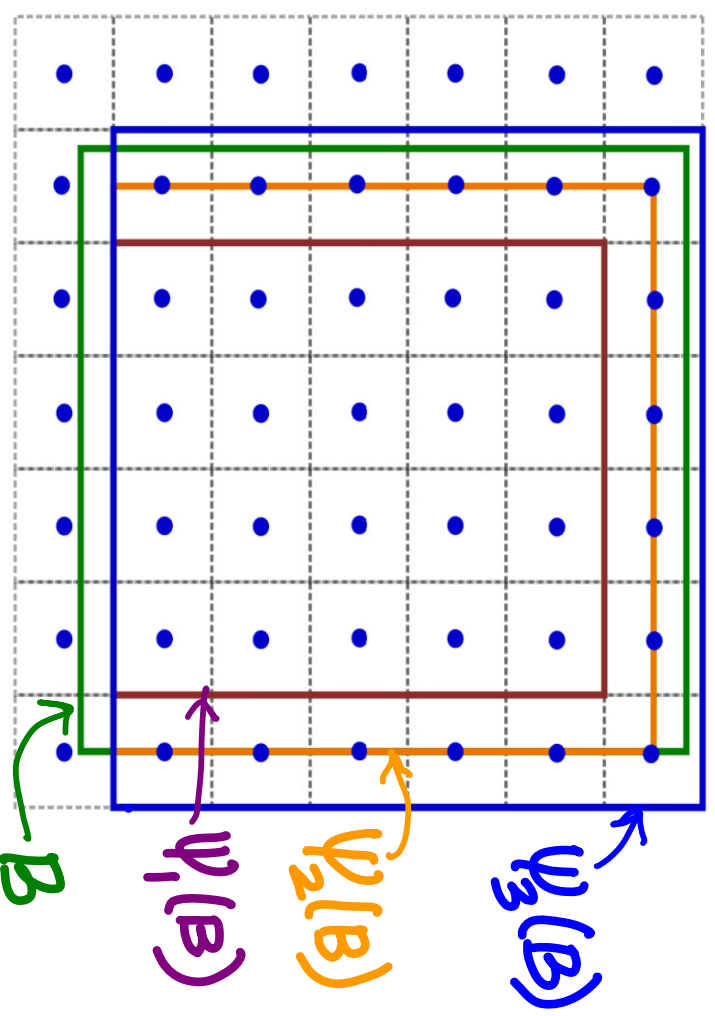
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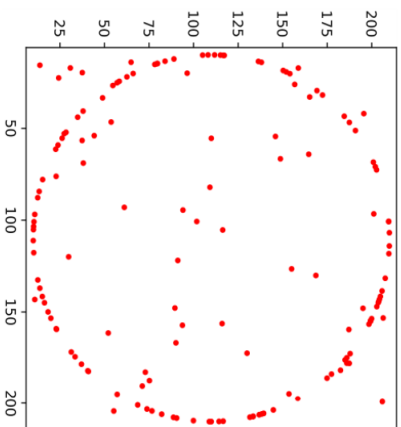
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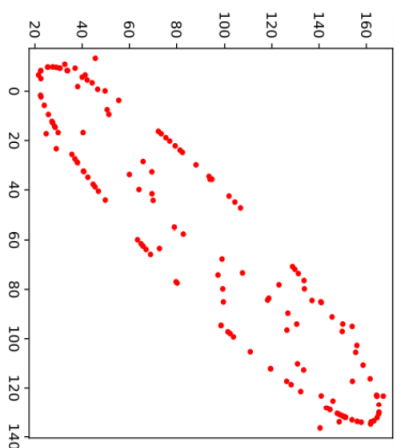


✓ analogous results (stability, etc.) to point cover

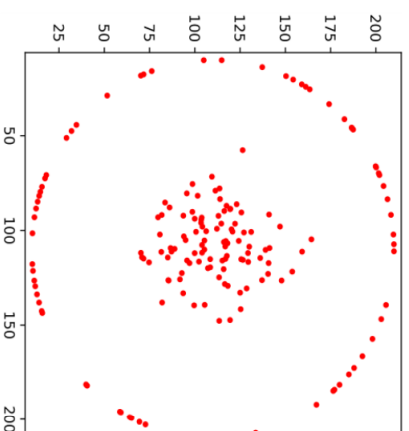
USE IN ML PIPELINES?



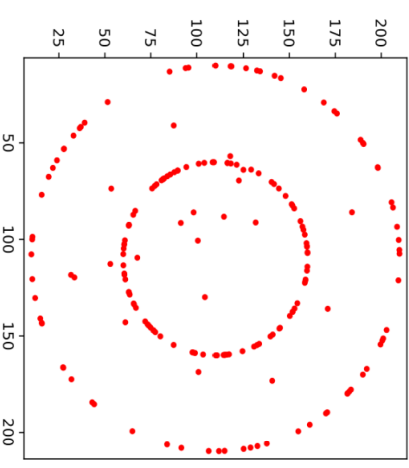
circle w/
noise
100 pts → 50 pts



ellipse w/
noise
100 pts → 50 pts

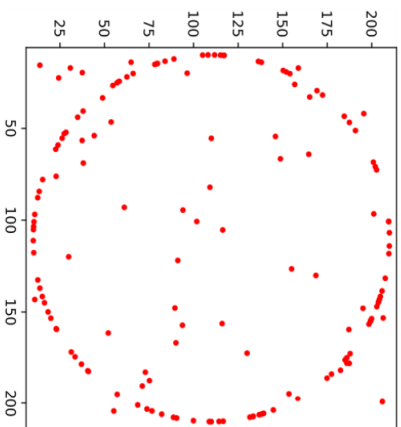


circle w/
dense cluster
75 pts → 100 pts

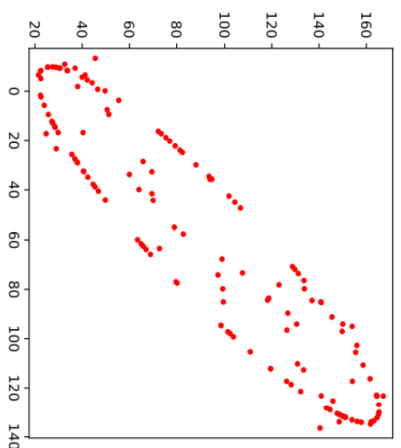


two circles
w/ noise
inner ○ : 100 pts
outer ○ : 75 pts
noise : 20 pts

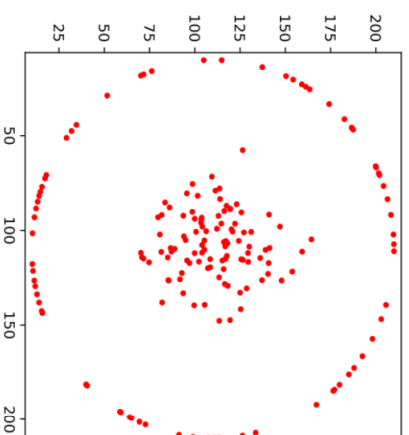
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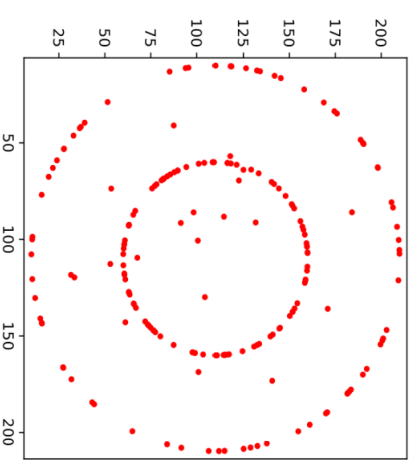
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ellipse w/
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circle w/
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* BF identifies feature(s) (circle(s), ellipse)
in each case for multiple & values

USE IN ML PIPELINES?

* data + noise (ϕ , low, medium) (4x3=12 PCs)

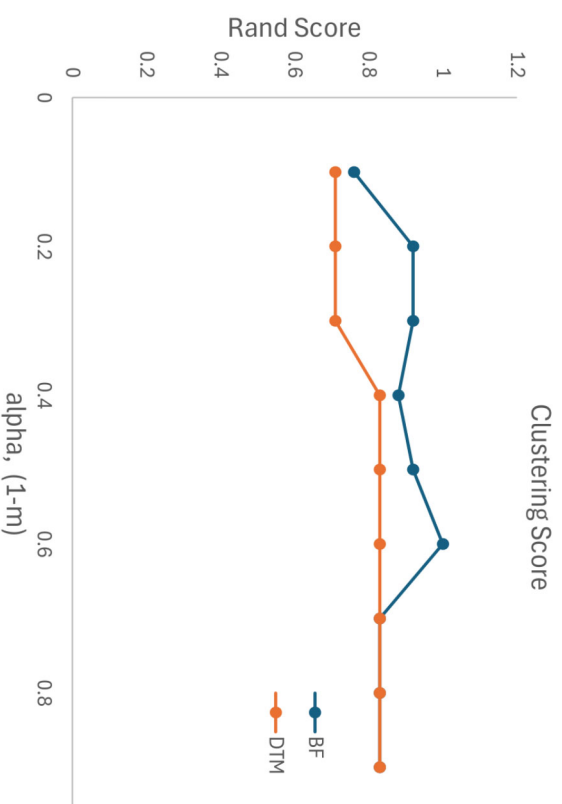
USE IN ML PIPELINES?

- * data + noise ($\emptyset$, low, medium) (4x3=12 PCs)
- * bottleneck dist. between PDs $\rightarrow$ clustering

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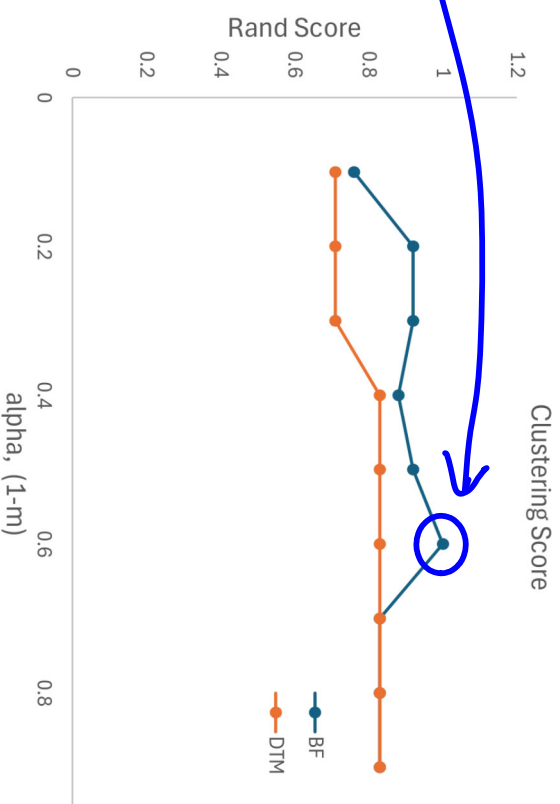


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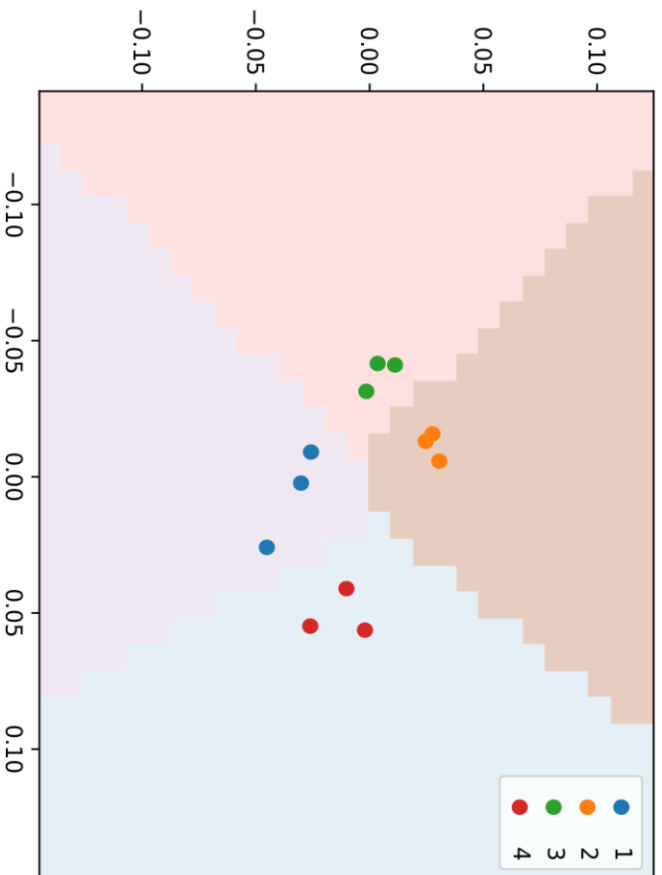
* Rand score

(=1 $\rightarrow$ perfect separation of clusters)

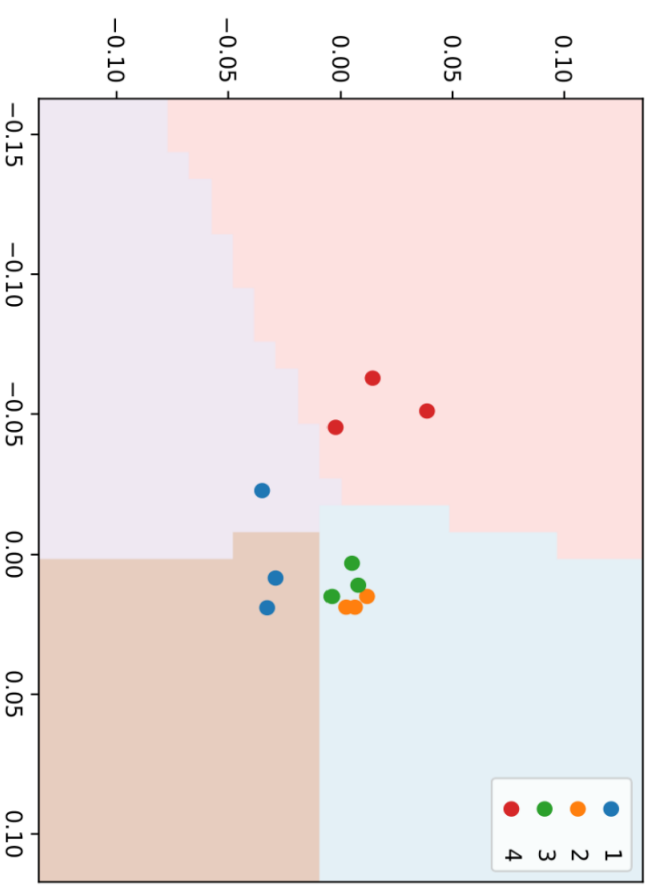


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- * bottleneck dist. between PDs $\rightarrow$ clustering



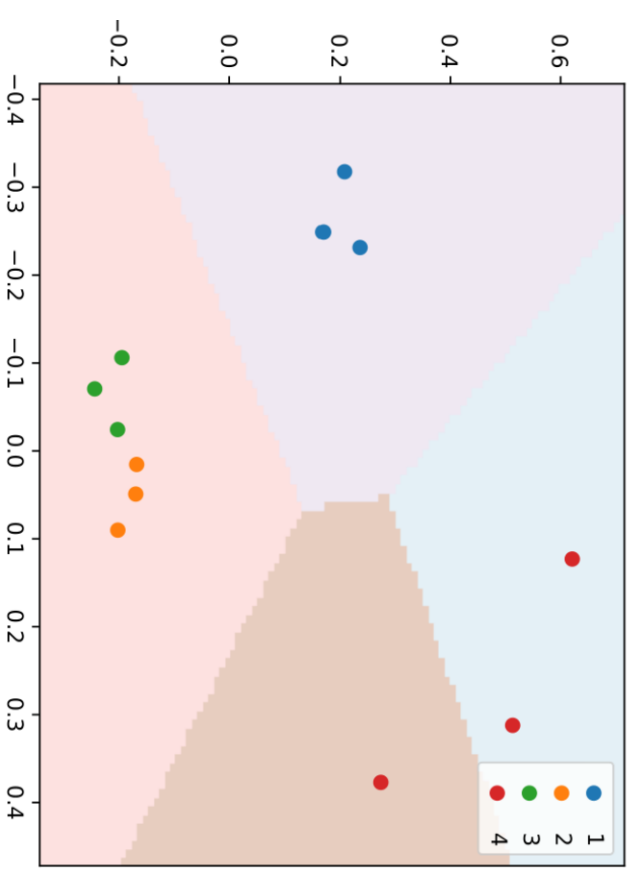
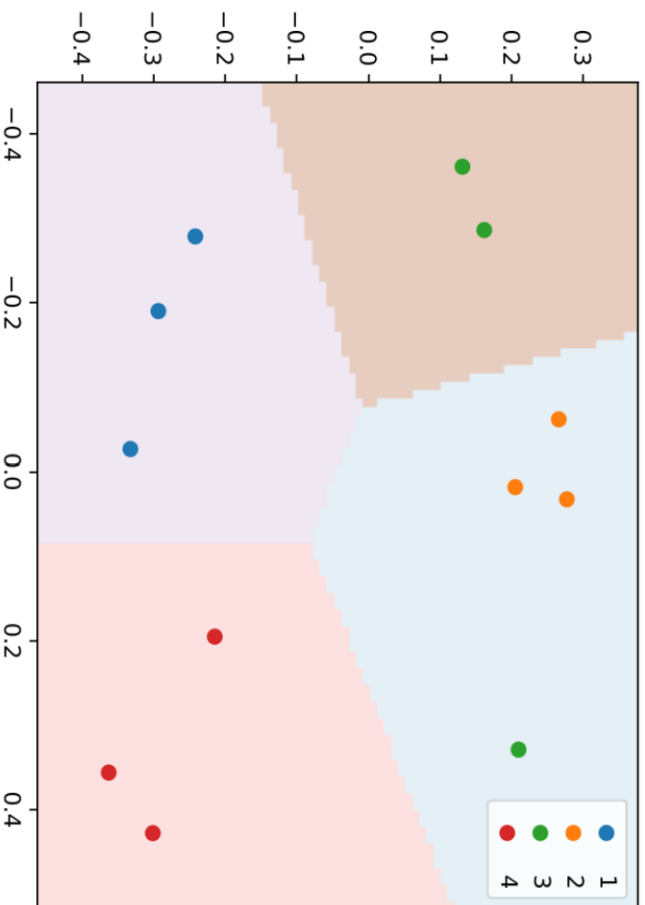
BF ($\alpha=0.6$)



DTM ($m=0.6$)

USE IN ML PIPELINES?

- * data + noise (ϕ , low, medium) (4x3=12 PCs)
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$$BF \left(\sum_i \text{dist}_i \right)$$

$$DTM \left(\sum_m \text{dist}_m \right)$$

BOX MAPPER

* (PCA+) k-Means clustering

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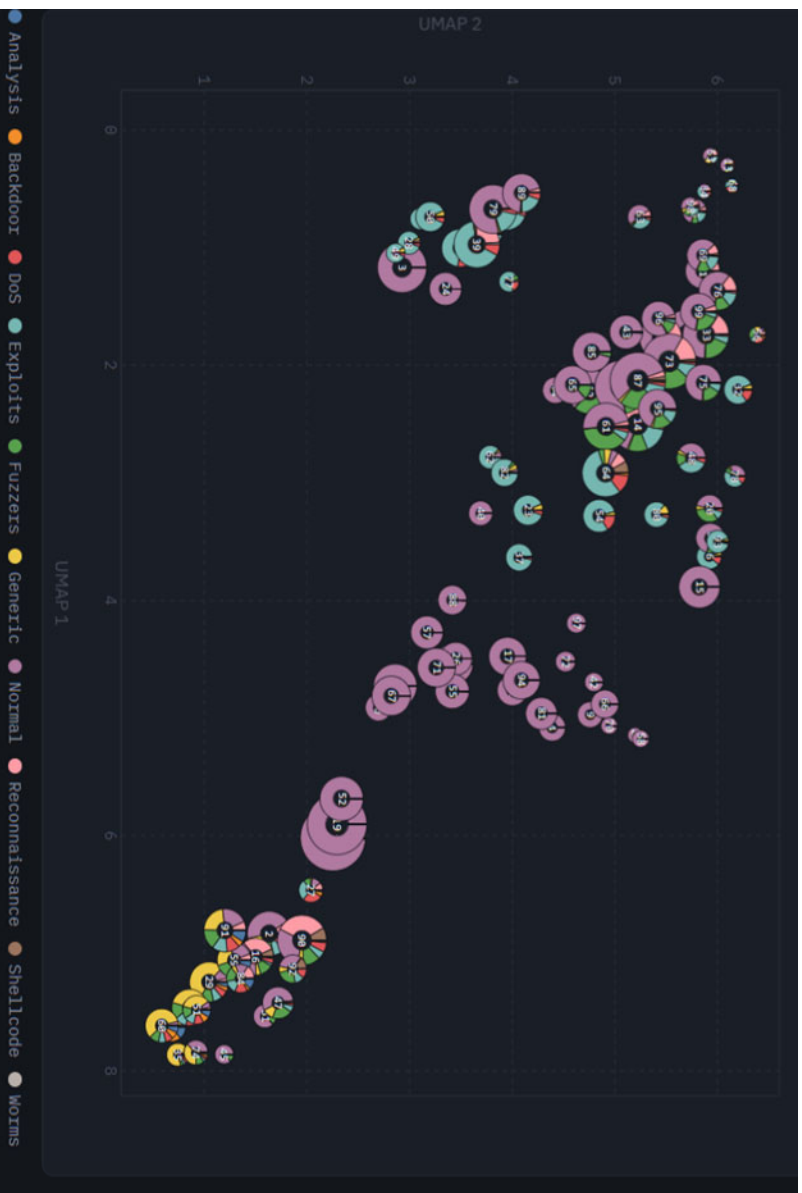
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 - 43,000 "pts" in 39 dim
 - normal + 9 attack categories
- ✓ 360s → 27s (~13x speed up)

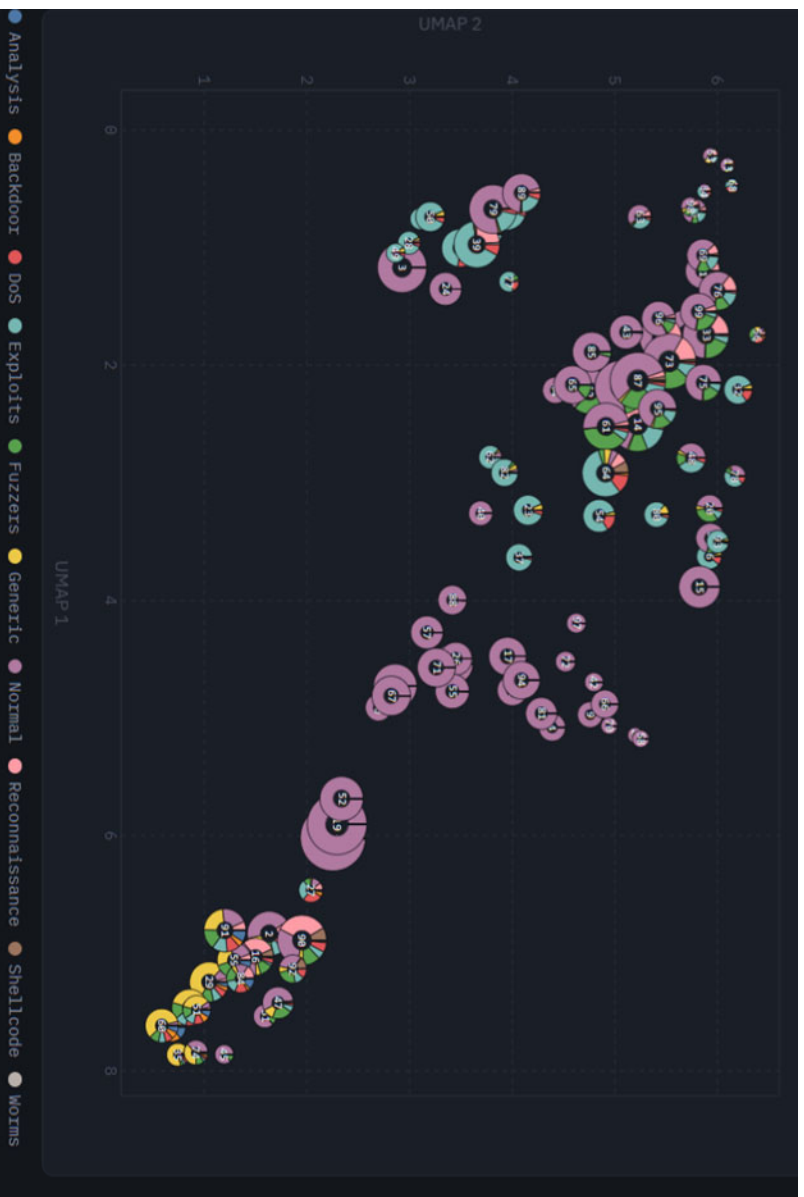
BOX MAPPER

box mapper w/
 $k=100$ clusters

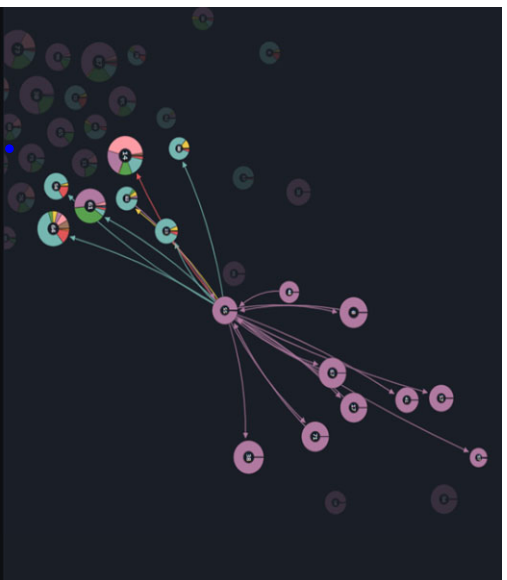


BOX MAPPER

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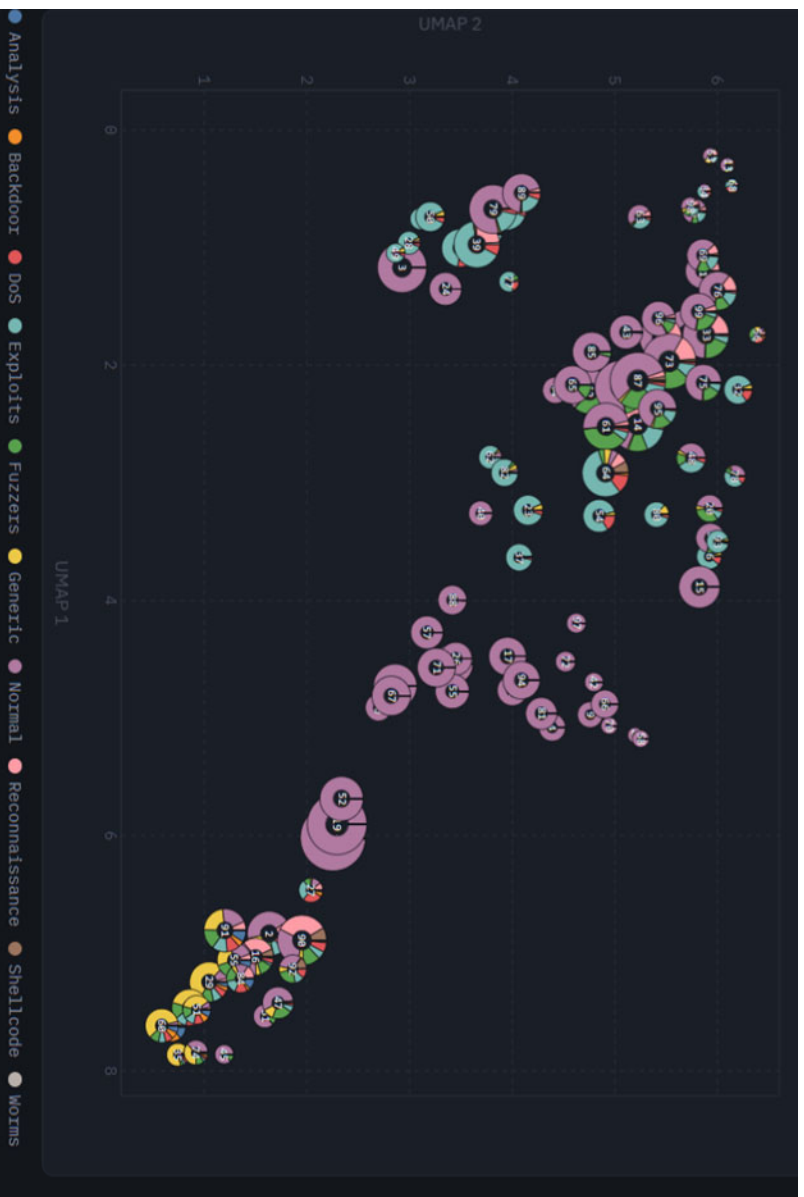


N'brs of
Node 55
(normal)



BOX MAPPER

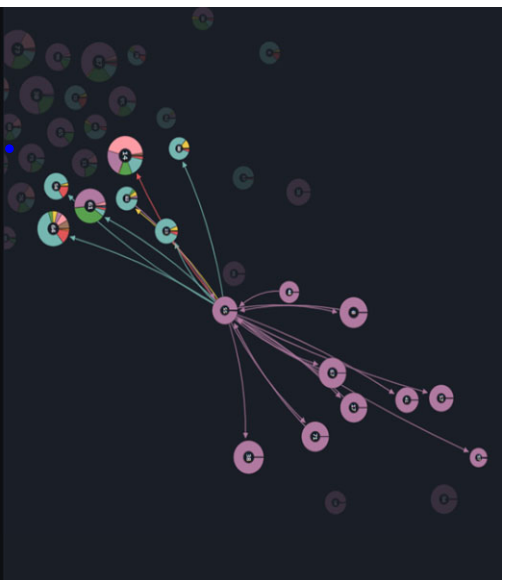
box mapper w/
 $k=100$ clusters



N'brs using
Euclidean distance

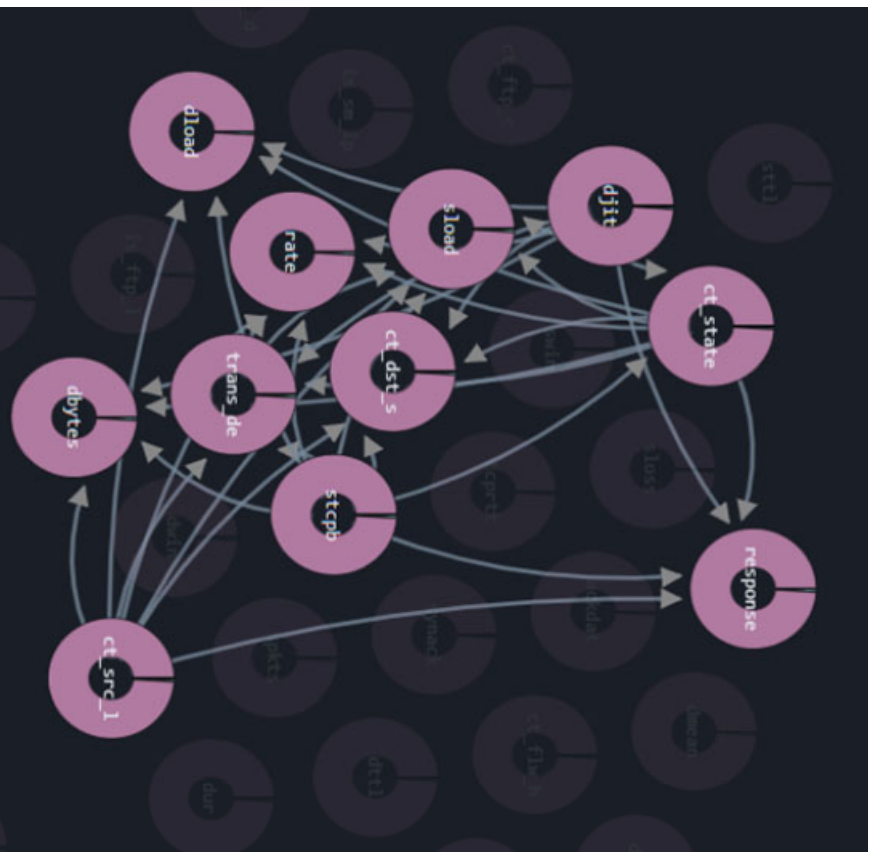


N'brs of
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BOX MAPPER

Feature spaces of Node 55 v/s Node 64 (anomalous)



32 edges



103 edges

POINTS TO NOTE

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Thank you!